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# 5. WATER QUALITY

## 5.1 Introduction

5.1.1.1 This section presents an assessment of potential water quality impacts arising from construction and operation of the Project, which has been conducted in accordance with the criteria and guidelines for evaluating and assessing water pollution as stated in Annex 6 and Annex 14 of the "*Technical Memorandum on Environmental Impact Assessment Process*" (EIAO-TM) as well as the requirements given in Clause 3.5.5 and Appendix D of the EIA Study Brief (No. ESB-373/2025) (hereinafter "the Study Brief").

## 5.2 Environmental Legislations, Standards and Guidelines

Technical Memorandum on Environmental Impact Assessment Ordinance (EIAO-TM)

5.2.1.1 The EIAO-TM was issued by Environmental Protection Department (EPD) under Section 16 of the EIAO. Reference sections in the EIAO-TM provide the details of the assessment criteria and guidelines that are relevant to the water quality assessment, including:

- Annex 6 - Criteria for Evaluating Water Pollution
- Annex 14 - Guidelines for Assessment of Water Pollution

Water Pollution Control Ordinance (WPCO)

5.2.1.2 The WPCO provides the major statutory framework for the protection and control of water quality in Hong Kong. According to the Ordinance and its subsidiary legislation, Hong Kong waters are divided into ten Water Control Zones (WCZs). Corresponding statements of Water Quality Objectives (WQOs) are stipulated for different water regimes (marine waters, inland waters, bathing beaches subzones, secondary contact recreation subzones and fish culture subzones) in the WCZs based on their beneficial uses. The Project falls within the Deep Bay WCZ as designated under the WPCO (Cap. 358). The WQOs for the watercourses in the Deep Bay WCZ relevant to the water quality impact assessment are listed in **Table 5.1**.

**Table 5.1 Summary of Key Water Quality Objectives for Deep Bay WCZ**

| Parameters | Objectives | Subzone |
|---|---|---|
| Aesthetic appearance | Waste discharges shall cause no objectionable odours or discolouration of the water. | Whole Zone |
| | Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent. | |
| | Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam. | |
| | There should be no recognisable sewage-derived debris. | |

| Parameters | Objectives | Subzone |
|---|---|---|
| | Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent. | |
| | Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits. | |
| Bacteria | The level of Escherichia coli should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year. | Secondary Contact Recreation Subzone and Mariculture Subzone |
| | The level of Escherichia coli should be zero per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days. | Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones |
| | The level of Escherichia coli should not exceed 1000 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days. | Yuen Long & Kam Tin (Lower) Subzone and other inland waters |
| | The level of Escherichia coli should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive in one calendar year. Samples should be taken at least 3 times in a calendar month at intervals of between 3 and 14 days. | Yung Long Bathing Beach Subzone |
| Colour | Waste discharges shall not cause the colour of water to exceed 30 Hazen units. | Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones |
| | Waste discharges shall not cause the colour of water to exceed 50 Hazen units. | Yuen Long & Kam Tin (Lower) Subzone and other inland waters |
| Dissolved Oxygen | Waste discharges shall not cause the level of DO to fall below 4 mg/L for 90% of the sampling occasions during the year, values should be taken at 1 metre below surface | Inner Marine Subzone excepting Mariculture Subzone |
| | Waste discharges shall not cause the level of DO to fall below 4 mg/L for 90% of the sampling occasions during the year, calculated as water column average. In addition, the concentration of DO should not be less than 2 mg/L within 2 m of the seabed for 90% of the sampling occasions during the year. | Outer Marine Subzone excepting Mariculture Subzone |
| | The DO level should not be less than 5 mg/L for 90% of the sampling occasions during the year, values should be taken at 1 metre below surface | Mariculture Subzone |
| | Waste discharges shall not cause the level of DO to be less than 4 mg/L | Yuen Long & Kam Tin (Upper and Lower) |

| Parameters | Objectives | Subzone |
|---|---|---|
| | | Subzones, Beas Subzone, Indus Subzone, Ganges Subzone, Water Gathering Ground Subzones and other inland waters of the Zone |
| pH | The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units | Marine waters excepting Yung Long Bathing Beach Subzone |
| | Waste discharges shall not cause the pH of water to exceed the range of 6.5-8.5 units. | Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones |
| | The pH of the water should be within the range of 6.0-9.0 units | Other inland waters |
| | The pH of the water should be within the range of 6.0-9.0 for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units | Yung Long Bathing Beach Subzone |
| Temperature | Waste discharges should not cause the natural daily temperature range to change by more than 2.0℃. | Whole zone |
| Salinity | Waste discharges shall not cause the natural ambient salinity level to change by more than 10%. | Whole zone |
| Suspended Solids (SS) | Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of SS which may adversely affect aquatic communities. | Marine waters |
| | Waste discharges shall not cause the annual median of SS to exceed 20 mg/L. | Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Ganges Subzone, Indus Subzone, Water Gathering Ground Subzones and other inland waters |
| Ammonia | The un-ionized ammoniacal nitrogen level should not be more than 0.021 mg/L, calculated as the annual average (arithmetic mean). | Whole zone |
| Nutrients | (a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants. | Inner and Outer Marine Subzones |
| | (b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.7 mg/L, expressed as annual mean. | Inner Marine Subzones |
| | (c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 mg/L, | Outer Marine Subzones |

| Parameters | Objectives | Subzone |
|---|---|---|
| | expressed as annual water column average. | |
| 5-day Biochemical Oxygen Demand ($BOD_5$) | Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 mg/L | Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones |
| | Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 mg/L | Yuen Long & Kam Tin (Lower) Subzone and other inland waters |
| Chemical Oxygen Demand (COD) | Waste discharges shall not cause the chemical oxygen demand to exceed 15 mg/L | Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones |
| | Waste discharges shall not cause the chemical oxygen demand to exceed 30 mg/L | Yuen Long & Kam Tin (Lower) Subzone and other inland waters |
| Toxins | Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic, carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other | Whole zone |
| | Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment. | Whole zone |
| Phenol | Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 mg/L as $C_6H_5OH$. | Yung Long Bathing Beach Subzone |
| Turbidity | Waste discharges shall not reduce light transmission substantially from the normal level. | Yung Long Bathing Beach Subzone |

*Source: Statement of Water Quality Objectives (Deep Bay Water Control Zone).*

Technical Memorandum for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS)

5.2.1.3 Discharge of effluents is subject to control under the WPCO. The TM-DSS gives guidance on the permissible effluent discharges based on the type of receiving waters (foul sewers, storm water drains, inland and coastal waters). The standards control the physical, chemical and microbial quality of effluents. The discharge limits vary with the effluent flow rates and the effluent should comply with the standards for effluent discharged into inland waters for irrigation, pond fish culture or amenity subject to the exact locations. The TM-DSS also specifies the discharge standards into foul sewers leading into the Government's sewage treatment plants. Subject to the flow rate of the effluents, corresponding standards for the effluent discharge into the Government's foul sewers should be followed.

Professional Persons Environmental Consultative Committee Practice Notes (ProPECC PNs)

5.2.1.4 A ProPECC PN was issued by the EPD to provide guidelines for handling and disposal of construction site discharges. The ProPECC PN 2/24 "*Construction Site Drainage*" provides good practice guidelines for dealing with 10 types of discharge from construction sites. These include surface run-off, groundwater, boring and drilling water, wastewater from concrete batching and/or precast concrete casting, wheel washing water, bentonite slurry, water for testing and/or sterilisation of water retaining structures and water pipes, wastewater from building constructions, acid cleaning, etching and pickling wastewater, and wastewater from site facilities. Practices given in the ProPECC PN 2/24 should be followed as far as possible during construction to minimise the water quality impact due to construction site drainage.

5.2.1.5 The ProPECC PN 1/23 "*Drainage Plans Subject to Comment by the EPD*" provides guidelines and practices for handling, treatment and disposal of various effluent discharges to stormwater drains and foul sewers during the operation phase. The design of site drainage and disposal of various site effluents generated within the new development area should follow the relevant guidelines and practices as given in the ProPECC PN 1/23.

Environment, Transport and Works Bureau Technical Circular (ETWB TC)

5.2.1.6 ETWB TC (Works) No. 5/2005 "*Protection of natural streams/rivers from adverse impacts arising from construction works*" provides an administrative framework to better protect all natural streams/rivers from the impacts of construction works. The procedures promulgated under this Circular aim to clarify and strengthen existing measures for protection of natural streams/rivers from government projects and private developments. The guidelines and precautionary mitigation measures given in the ETWB TC (Works) No. 5/2005 should be followed as far as possible to protect the inland watercourse at or near the Project area during construction phase.

Hong Kong Planning Standards and Guidelines (HKPSG)

5.2.1.7 Chapter 9 of the HKPSG outlines environmental requirements that need to be considered in land use planning. The recommended guidelines, standards and guidance cover the selection of suitable locations for the developments and sensitive uses, provision of environmental facilities, and design, layout, phasing and operational controls to minimise adverse environmental impacts. It also lists out environmental factors that influence land use planning and recommends buffer distances for land uses.

## 5.3 Description of Environment

5.3.1.1 The Project falls within the Deep Bay WCZ as designated under the WPCO. According to Item 3.5.5.2 of the EIA Study Brief (No. ESB-373/2025), the assessment area for this water quality impact assessment should include areas within 500 metres from the boundary of the Project and its associated works. The baseline condition of water bodies in the assessment area was established based on river water quality monitoring data collected by EPD and Drainage Services Department (DSD). Descriptions of the baseline conditions provided in the subsequent sections are extracted from EPD's report "*River Water Quality in Hong*

*Kong in 2024*" which contain the latest information on river water quality respectively at the time of preparing this EIA Report.

5.3.1.2 The assessment area lies within the Kam Tin River and River Beas catchment area. The water quality monitoring results at EPD's river water quality monitoring stations at Kam Tin River (KT1 and KT2) and River Beas (RB1, RB2 and RB3) are shown in **Table 5.2** and **Table 5.3** respectively. The locations of these stations are illustrated in **Figure 5.1**. For Kam Tin River, the overall WQO compliance rate in 2024 was 53%, as compared with 21% in 1994. Both monitoring stations (KT1 and KT2) at Kam Tin River were graded "Fair" and "Bad" respectively in 2024. For River Beas, the overall WQO compliance rate was 81% in 2024, as compared with 27% in 1994. Its three monitoring stations (RB1, RB2 and RB3) all achieved "Good" WQI gradings in 2024.

**Table 5.2 Summary Statistics of River Water Quality of Kam Tin River Collected by EPD in 2024**

| Parameters | Kam Tin River | | WPCO WQO |
|---|---|---|---|
| | KT1 | KT2 | (Yuen Long & Kam Tin (Upper) Subzone) |
| Dissolved Oxygen (DO) (mg/L) | 6.3 (4.4 - 7.5) | 3.9 (1.7 - 5.8) | $\geqslant$ 4 mg/L |
| pH | 7.5 (7.1 - 8.3) | 7.4 (7.2 - 7.8) | within 6.5 - 8.5 units |
| Suspended Solids (SS) (mg/L) | 9.7 (1.6 - 30.0) | 14.0 (2.7 - 73.0) | annual median: $\leqslant$ 20 mg/L |
| 5-day Biochemical Oxygen Demand ($BOD_5$) (mg/L) | 9.1 (4.8 - 19.0) | 16.5 (8.0 - 93.0) | $\leqslant$ 3 mg/L |
| Chemical Oxygen Demand (COD) (mg/L) | 13 (3 - 43) | 28 (4 - 170) | $\leqslant$ 15 mg/L |
| Oil & Grease (mg/L) | <0.5 (<0.5 - 0.7) | 0.7 (<0.5 - 1.9) | Not available |
| *E. coli* (cfu/100mL) | 33 324 (14 000 - 210 000) | 81 012 (22 000 - 330 000) | Not available |
| Faecal Coliforms (cfu/100mL) | 192 045 (80 000 - 360 000) | 328 299 (140 000 - 1 000 000) | Not available |
| Ammonia Nitrogen (mg/L) | 4.150 (0.980 - 6.400) | 7.450 (1.300 - 13.000) | Not available |
| Nitrate Nitrogen (mg/L) | 1.150 (0.430 - 1.500) | 0.380 (<0.003 - 1.300) | Not available |
| Total Kjeldahl Nitrogen (mg/L) | 5.25 (1.30 - 8.50) | 9.30 (1.30 - 19.00) | Not available |
| Orthophosphate Phosphorus ($PO_4$-P) (mg/L) | 0.615 (0.340 - 1.100) | 0.815 (0.380 - 3.900) | Not available |
| Total Phosphorus (mg/L) | 0.86 (0.38 - 1.50) | 1.30 (0.50 - 4.90) | Not available |

| Parameters | Kam Tin River | | WPCO WQO (Yuen Long & Kam Tin (Upper) Subzone) |
|---|---|---|---|
| | KT1 | KT2 | |
| Sulphide (mg/L) | <0.02 (<0.02 - 0.03) | 0.03 (<0.02 - 0.32) | Not available |
| Aluminium (Al) (µg/L) | <50 (<50 - 240) | <50 (<50 - <50) | Not available |
| Cadmium (Cd) (µg/L) | <0.1 (<0.1 - <0.1) | <0.1 (<0.1 - <0.1) | Not available |
| Chromium (Cr) (µg/L) | <1 (<1 - 2) | <1 (<1 - <1) | Not available |
| Copper (Cu) (µg/L) | 2 (<1 - 4) | 2 (<1 - 4) | Not available |
| Lead (Pb) (µg/L) | <1 (<1 - <1) | <1 (<1 - <1) | Not available |
| Zinc (Zn) (µg/L) | <10 (<10 - 20) | <10 (<10 - 40) | Not available |
| Flow ($m^3/s$) | 0.506 (0.135 - 1.274) | 0.473 (0.172 - 1.132) | Not available |

*Note:*1. Data source: EPD River Water Quality in Hong Kong in 2024.
2. Data presented are in annual medians of monthly samples; except those for faecal coliforms and E. coli which are in annual geometric means.
3. Figures in brackets are annual ranges.
4. NM indicates no measurement taken.
5. cfu – colony forming unit.
6. Values at or below laboratory reporting limits are presented as laboratory reporting limits.
7. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

## Table 5.3 Summary Statistics of River Water Quality of River Beas Collected by EPD in 2024

| Parameters | River Beas | | | WPCO WQO (Beas Subzone) |
|---|---|---|---|---|
| | RB1 | RB2 | RB3 | |
| Dissolved Oxygen (DO) (mg/L) | 9.0 (8.4 - 11.1) | 7.4 (5.6 - 8.3) | 6.1 (4.0 - 9.3) | ⩾ 4 mg/L |
| pH | 7.6 (7.3 - 8.2) | 7.1 (6.7 - 7.3) | 7.1 (6.8 - 7.7) | within 6.5 - 8.5 units |
| Suspended Solids (SS) (mg/L) | 4.5 (<1.0 - 13.0) | 9.1 (1.6 - 110.0) | 9.7 (0.8 - 160.0) | annual median: ⩽ 20 mg/L |
| 5-day Biochemical Oxygen Demand ($BOD_5$) (mg/L) | 1.8 (1.0 - 15.0) | 4.5 (2.0 - 17.0) | 5.8 (2.7 - 23.0) | ⩽ 3 mg/L |
| Chemical Oxygen Demand (COD) (mg/L) | 7 (3 - 18) | 12 (5 - 65) | 16 (4 - 26) | ⩽ 15 mg/L |

| Parameters | River Beas | | | WPCO WQO (Beas Subzone) |
|---|---|---|---|---|
| | RB1 | RB2 | RB3 | |
| Oil & Grease (mg/L) | <0.5 (<0.5 - <0.5) | <0.5 (<0.5 - 1.1) | <0.5 (<0.5 - <0.5) | Not available |
| *E. coli* (cfu/100mL) | 4 392 (900 - 38 000) | 33 427 (6 600 - 460 000) | 15 316 (3 300 - 80 000) | Not available |
| Faecal Coliforms (cfu/100mL) | 17 947 (1 900 - 180 000) | 103 039 (39 000 - 470 000) | 82 181 (8 500 - 490 000) | Not available |
| Ammonia Nitrogen (mg/L) | 0.130 (0.060 - 0.420) | 0.785 (0.170 - 4.400) | 1.100 (0.320 - 4.400) | Not available |
| Nitrate Nitrogen (mg/L) | 1.100 (0.480 - 1.600) | 0.620 (<0.005 - 1.400) | 0.690 (0.250 - 1.900) | Not available |
| Total Kjeldahl Nitrogen (mg/L) | 0.32 (0.17 - 1.10) | 1.20 (0.30 - 7.50) | 1.85 (0.46 - 5.70) | Not available |
| Orthophosphate Phosphorus ($PO_4$-P) (mg/L) | 0.230 (0.110 - 0.380) | 0.175 (0.089 - 0.890) | 0.150 (0.073 - 0.330) | Not available |
| Total Phosphorus (mg/L) | 0.30 (0.17 - 0.39) | 0.23 (0.17 - 1.30) | 0.28 (0.15 - 0.57) | Not available |
| Sulphide (mg/L) | <0.02 (<0.02 - <0.04) | <0.02 (<0.02 - 0.03) | <0.02 (<0.02 - 0.05) | Not available |
| Aluminium (Al) (µg/L) | <50 (<50 - 50) | <50 (<50 - <50) | <50 (<50 - 50) | Not available |
| Cadmium (Cd) (µg/L) | <0.1 (<0.1 - <0.1) | <0.1 (<0.1 - <0.1) | <0.1 (<0.1 - <0.1) | Not available |
| Chromium (Cr) (µg/L) | <1 (<1 - <1) | <1 (<1 - <1) | <1 (<1 - 1) | Not available |
| Copper (Cu) (µg/L) | 1 (1 - 2) | 1 (<1 - 2) | 1 (<1 - 2) | Not available |
| Lead (Pb) (µg/L) | <1 (<1 - <1) | <1 (<1 - <1) | <1 (<1 - <1) | Not available |
| Zinc (Zn) (µg/L) | <10 (<10 - 20) | <10 (<10 - 30) | <10 (<10 - 40) | Not available |
| Flow ($m^3/s$) | 0.108 (0.024 - 0.504) | 0.176 (0.007 - 0.594) | NM | Not available |

*Note:1. Data source: EPD River Water Quality in Hong Kong in 2024.*

2. *Data presented are in annual medians of monthly samples; except those for faecal coliforms and E. coli which are in annual geometric means.*
3. *Figures in brackets are annual ranges.*
4. *NM indicates no measurement taken.*
5. *cfu – colony forming unit.*
6. *Values at or below laboratory reporting limits are presented as laboratory reporting limits.*
7. *Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.*

5.3.1.3 In additional to the EPD's routine river water quality monitoring data, the water quality monitoring data collected by DSD at Yuen Long Bypass Floodway and Ngau Tam Mei Drainage Channel was also used to establish the baseline water quality conditions. The water quality monitoring data collected between year 2013 and year 2020 at the upstream and downstream stations of these two channels are tabulated in **Table 5.4** below.

**Table 5.4 Summary Statistics of River Water Quality of Yuen Long Bypass Floodway and Ngau Tam Mei Channel Collected by DSD**

| Parameters | Yuen Long Bypass Floodway (YLBF) | | Ngau Tam Mei Drainage Channel (NTMDC) | | WPCO WQO (inland waters) |
|---|---|---|---|---|---|
| | Upstream | Downstream | Upstream | Downstream | |
| Dissolved Oxygen (DO) (% Saturation) | 66 (44 - 107) | 60 (33 - 313) | 57 (30 - 95) | 62 (18 - 91) | Not available |
| pH | 8.0 (7.3 - 8.6) | 7.8 (7.3 - 8.5) | 7.5 (7.2 - 8.3) | 7.7 (7.2 - 8.1) | YLBF: 6.5 - 8.5; NTMDC: 6.0 - 9.0 |
| 5-day Biochemical Oxygen Demand ($BOD_5$) (mg/L) | 6.0 (2.0 - 8.5) | 5.3 (3.0 - 7.5) | 7 (4 - 11) | 6 (2 - 10) | YLBF: $\leqslant$ 3 mg/L NTMDC: $\leqslant$ 5 mg/L |
| Ammonia as N (as N mg/L) | 6.0 (0.9 - 23.1) | 6.3 (0.5 - 9.6) | 4.1 (0.5 - 7.7) | 4.5 (0.5 - 6.8) | Not available |

*Note:1. Data source: DSD's Eco-channels Database Management System (EcoDMS) (data collected between year 2013 and year 2020).*
*2. Data presented are medians.*
*3. Data in brackets indicate the ranges.*
*4. YLBF falls within the Yuen Long & Kam Tin (Upper) Subzone and NTMDC is identified as "other inland waters" in Deep Bay WCZ (see **Table 5.1** for their specified WQOs).*

5.3.1.4 Based on the monitoring results, low DO saturations were recorded at these two channels. High $BOD_5$ and ammonia-nitrogen levels indicated that these channels are contaminated by organic and nutrient pollutants.

## 5.4 Water Sensitive Receivers

5.4.1.1 Water Sensitive Receivers (WSRs) within the 500m assessment area were identified according to Item 3.5.5.2 of the EIA Study Brief (No. ESB-373/2025). Representative WSRs were identified based on review of relevant land use plans including Outline Zoning Plans (OZP), information available in the Statutory Planning Portal of the Town Planning Board (TPB) and land status plans published by Lands Department. These include Kam Tin River, Yuen Long Bypass Floodway, Ngau Tam Mei Drainage Channel, River Beas (Sheung Yue River), ponds (adjacent to Kam Tin River, within Ngau Tam Mei area, near Greenery Gardens, Pang Loon Tei, Ki Lun Tsuen, Cheung Lek and Tit Hang), streams (from hillsides of Ho Hok Shan, Kai Kung Leng, Ngau Tam Shan, Hadden Hill and Tit Hang Shan), Kwu Tung Reservoir, Lam Tsuen Country Park (LTCP), Wetland Conservation Area (WCA); and Wetland Buffer Area (WBA). Other potential WSRs such as inner Deep Bay and Mai Po Ramsar Site are located at further downstream of Kam Tin River which are considered far away from site (>3km compared to the 500m assessment area) and water quality impact is not anticipated.

5.4.1.2 Major WSRs within 500m from the Project site boundary were identified and are presented in **Table 5.5** and their respective locations are illustrated in **Figures 5.1.1 to 5.1.4**.

**Table 5.5 Water Sensitive Receivers**

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| *Watercourse* | | | | |
| W01 | Kam Tin River | modified watercourse | partially within Project Site | bridge piers will be constructed in Kam Tin River |
| W02 | Yuen Long Bypass Floodway | modified watercourse | adjacent to Project Site | - |
| W03 | Ngau Tam Mei Drainage Channel (NTMDC) | modified watercourse | 105 m | - |
| W04 | River Beas (Sheung Yue River) | modified watercourse | 230 m | - |
| W05 | hillside of Ho Hok Shan | natural watercourse | 330 m | - |
| W06 | Ko Po Tsuen | modified watercourse | 440 m | - |
| W07 | near Tung Shing Lei, along Yuen Long Highway | modified watercourse | partially within Project Site | will be retained |
| W08 | near Small Traders New Village | natural watercourse | 340 m | - |
| W09 | old Kam Tin River and its tributaries, south of Nam Sang Wai | natural watercourse | 300 m | - |
| W10 | old Kam Tin River, within West Rail Compensatory Wetland | natural watercourse | 330 m | - |
| W11 | old Kam Tin River, near Cheung Chun San Tsuen | natural watercourse | 155 m | - |
| W12 | old Kam Tin River, near Koon Chun Sauce Factory | natural watercourse | 55 m | - |
| W13 | Cheung Chun San Tsuen | modified watercourse | 365 m | - |
| W14 | tributary of Kam Tin River, near Park Yoho | modified watercourse | partially within Project Site | will be retained |
| W15 | tributary of Kam Tin River (downstream of W16) | modified watercourse | within Project Site | will be retained |
| W16 | near Sha Po Tsuen | modified watercourse | partially within Project Site | will be retained |

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| W17 | south of Koon Chun Sauce Factory | modified watercourse | adjacent to Project Site | - |
| W18 | near Construction Sector Imported Labour Quarters | modified watercourse | partially within Project Site | will be retained |
| W19 | drainage along San Tin Highway / Sam Tam Road / Castle Peak Road – Tam Mi | modified watercourse | partially within Project Site | will be retained |
| W20 | hillside of Kai Kung Leng | natural watercourse | partially within Project Site | will be retained |
| W21 | along Pok Wai Road / Pok Wai West Road | modified watercourse | partially within Project Site | will be retained |
| W22 | along Pok Wai Road / Kam Pok Road East | modified watercourse | 40 m | - |
| W23 | near Wah On Villa (downstream of W24) | modified watercourse | 110 m | - |
| W24 | near Wah On Villa (upstream of W23) | semi-natural watercourse | partially within Project Site | will be retained |
| W25 | near Ching Yau Road | modified watercourse | partially within Project Site | will be retained |
| W26 | tributaries of NTMDC | semi-natural watercourse | 310 m | - |
| W27 | near existing Ngau Tam Mei Ventilation Building (near Ching Yau Path) | modified watercourse | partially within Project Site | section within Project Site will be diverted and/or modified |
| W28 | tributaries of NTMDC (joins W27 before discharge to NTMDC) | semi-natural watercourse | partially within Project Site | will be retained |
| W29 | tributaries of NTMDC (joins W27 & W28 before discharge to NTMDC) | modified watercourse | 50 m | - |
| W30 | tributaries of NTMDC | semi-natural watercourse | 120 m | - |
| W31 | tributaries of NTMDC | semi-natural watercourse | partially within Project Site | section(s) at/near portal will be diverted and/or modified |
| W32 | tributaries of NTMDC (upstream section of W31) | modified watercourse | 150 m | - |
| W33 | near Ngau Tam Mei Water Treatment Works (upstream section of W32) | modified watercourse | partially within Project Site | will be retained |

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| W34 | tributaries of NTMDC | modified watercourse | 270 m | - |
| W35 | tributaries of NTMDC (upstream sections of W36) | modified watercourse | 270 m | - |
| W36 | tributaries of NTMDC | natural watercourse | 360 m | - |
| W37 | hillside of Ngau Tam Shan | natural watercourse | partially within Project Site | section(s) at/near portal will be diverted and/or modified |
| W38 | near San Tin Barracks | semi-natural watercourse | 330 m | - |
| W39 | Pang Loon Tei | semi-natural watercourse | partially within Project Site | will be retained |
| W40 | Ki Lun Tsuen | semi-natural watercourse | partially within Project Site | will be retained |
| W41 | Ki Lun Tsuen | semi-natural watercourse | partially within Project Site | will be retained |
| W42 | old River Beas | natural watercourse | 190 m | - |
| W43 | tributaries of River Beas | modified watercourse | 315 m | - |
| W44 | hillside of Hadden Hill | natural watercourse | partially within Project Site | section(s) at/near portal will be diverted and/or modified |
| W45 | Downstream channel of Kwu Tung Reservoir | modified watercourse | 215 m | - |
| W46 | near San Tin Interchange | modified watercourse | 320 m | - |
| W47 | drainage along Fanling Highway / Kwu Tung Road / Castle Peak Road – Chau Tau | modified watercourse | partially within Project Site | will be retained |
| W48 | Chau Tau Tsuen | modified watercourse | 150 m | - |
| W49 | Chau Tau Tsuen | modified watercourse | partially within Project Site | will be retained |
| W50 | drainage along Fanling Highway / Kwu Tung Road / Castle Peak Road – Chau Tau | modified watercourse | partially within Project Site | will be retained |
| W51 | Kwu Tung North New Development Area | modified watercourse | partially within Project Site | will be retained |
| W52 | Tung Fong | modified watercourse | 440 m | - |

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| W53 | hillside of Mount Kirkpatrick | natural watercourse | partially within Project Site | will be retained |
| W54 | hillside of Tit Hang Shan | natural watercourse | 135 m | - |
| W55 | Tit Hang | modified watercourse | 190 m | - |
| *Ponds* | | | | |
| P01 | Nam Sang Wai | ponds | adjacent to Project Site | - |
| P02 | Tai Sang Wai | ponds | 160 m | - |
| P03 | near Small Traders New Village | pond | 290 m | - |
| P04 | near Tung Shing Lei | ponds | 45 m | - |
| P05 | adjacent to old Kam Tin River (W11), near Cheung Chun San Tsuen | ponds | 275 m | - |
| P06 | adjacent to old Kam Tin River (W11), near Cheung Chun San Tsuen | ponds | 350 m | to be removed (by other project) |
| P07 | AFCD Au Tau Fisheries Office | ponds | 130 m | - |
| P08 | adjacent to old Kam Tin River (W12) | ponds | 155 m | - |
| P09 | north of Park Yoho | ponds | 5 m | - |
| P10 | south of Sha Po Sewage Pumping Station | ponds | 10 m | - |
| P11 | Sha Po Tsuen | pond | 370 m | - |
| P12 | near Wing Kei Tsuen | ponds | 110 m | - |
| P13 | Pok Wai Flood Storage Ponds | ponds | 180 m | - |
| P14 | adjacent to Kam Pok Road East | pond | 145 m | - |
| P15 | between Kam Pok Road East and Fairview Park Boulevard | ponds | 235 m | - |
| P16 | near Wah On Villa | ponds | 40 m | - |
| P17 | along Ngau Tam Mei Drainage Channel (W03) in Ngau Tam Mei Area | ponds | partially within Project Site | to be removed (by other project) and partially enhanced by this Project |
| P18 | Ngau Tam Mei | pond | 20 m | - |
| P19 | Ngau Tam Mei, adjacent to natural stream running from | ponds | 40 m | - |

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| | hillslope of Kai Kung Leng | | | |
| P20 | near Ngau Tam Mei Water Treatment Works | ponds | within Project Site | will be retained |
| P20a | near Ngau Tam Mei Water Treatment Works | ponds | within Project Site | ponds will be enhanced |
| P21 | west of San Tin Barracks | pond | 495 m | to be removed (by other project) |
| P22 | south of Pang Loon Tei | ponds | 395 m | to be removed (by other project) |
| P23 | east of Pang Loon Tei | ponds | adjacent to Project Site | to be removed (by other project) |
| P24 | south of Ki Lun Tsuen Playground | ponds | within Project Site | will be retained |
| P24a | south of Ki Lun Tsuen Playground | ponds | within Project Site | ponds will be removed |
| P25 | Ki Lun Tsuen, along River Beas | pond | 320 m | - |
| P26 | Ki Lun Tsuen, northwest of Cheung Lek | ponds | 125 m | - |
| P27 | Kwu Tung Reservoir | Reservoir | 95 m | - |
| P28 | adjacent to downstream channel of Kwu Tung Reservoir | pond | 260 m | - |
| P29 | Tit Hang | pond | 270 m | - |
| *Others* | | | | |
| - | Wetland Conservation Area (WCA) | Conservation Area | partially within Project Site | Project Site within WCA near Nam Sang Wai |
| - | Wetland Buffer Area (WBA) | Conservation Area | partially within Project Site | Project Site within WBA near Nam Sang Wai |
| - | Conservation Area (CA) | Conservation Area | partially within Project Site | piers will be constructed in CA near Wah On Villa; portal will be constructed in CA near Ngau Tam Mei Water Treatment Works |
| - | West Rail Compensatory Wetlands | Conservation Area | 325 m | - |
| - | Yuen Long Bypass Floodway Engineered Wetland | Conservation Area | 30 m | - |

| ID | Description | Nature | Approximate Distance to Project Site | Remarks |
|---|---|---|---|---|
| - | Lam Tsuen Country Park (LTCP) | Country Park | adjacent to Project Site | - |

5.4.1.3 W01 is a channelized section of Kam Tin River near Nam Sang Wai. Kam Tin River is the largest drainage system in the eastern region of Yuen Long. The mainstream of Kam Tin River is approximately 13 km long, originates from Tai Mo Shan, flows from southeast to northwest, passes through areas in Lui Kung Tin, Kap Lung, Tai Ling, Shek Kong, and eventually converges with Shan Pui River near Lut Chau before discharge to inner Deep Bay. The catchment area of Kam Tin River covers about 44.3 $km^2$, which is bounded by the watersheds of Tai To Yan to the east, western ridge of Tai Mo Shan to the south, Ho Hok Shan (W05) to the west and Kai Kung Leng (W20) to the north.

5.4.1.4 Yuen Long Bypass Floodway (W02) is constructed to mitigate flooding in Yuen Long Town and its peripheral village areas. It is a 3.8 km long with over 20 m wide large drainage channel built at the South of Yuen Long Town to intercept 40% of the runoff in the Yuen Long catchment. Running from southwest to northeast, the intercepted flow is diverted to the downstream of Kam Tin River and then discharged to Deep Bay.

5.4.1.5 Ngau Tam Mei Drainage Channel (W03) is a drainage channel, running from east to west, covers a catchment area of about 10 $km^2$, which is bounded by the Kai Kung Leng (W20) to the south and Ngau Tam Shan (W37) to the north. In the past, the low-lying Ngau Tam Mei area was classified as a major flooding black spot. To alleviate the severe flooding problem, a total of 4 km long Ngau Tam Mei Drainage Channel (NTMDC) was built to divert the flood water from the Ngau Tam Mei area to Kam Tin River. Construction of the channel was completed in 2005 and the flooding problem at Ngau Tam Mei area is now resolved.

5.4.1.6 W04 is a channelized section of River Beas (W04, also known as Sheung Yue River) near Ki Lun Tsuen. It originates from Tai To Yan, runs from south to north, collects flows from Kai Kung Leng, Ngau Tam Shan (W37) and Hadden Hill (W44), passes through Kwu Tung, and eventually joins with Shek Sheung River and River Indus near Ho Sheung Heung. In the 1990s, the areas along River Beas were frequently experiencing severe flooding problem during rainy season. In 2001, the river was trained by straightening the river alignment to improve the flow path as well as widening the river width to increase its hydraulic capacity.

5.4.1.7 Several natural watercourses at the hillside of Ho Hok Shan (W05), Kai Kung Leng (W20), Ngau Tam Shan (W37), Hadden Hill (W44), Mount Kirkpatrick (W53) and Tit Hang Shan (W54) are identified within the assessment area. Although partial sections of these watercourses (i.e. W20, W37, W44 and W53) are located within the project site boundary, only the natural watercourses at Ngau Tam Shan near the proposed Ngau Tam Mei Interchange, T-junction in Ki Lun Tsuen (east of Saddle Pass) and Hadden Hill (i.e. W37 and W44) would be affected by construction of the Project.

5.4.1.8 There are several natural (W08 to W12) and modified (W06, W07, W13 to W18) watercourses identified along the Kam Tin River. The natural watercourses are the meanders of old Kam Tin River and its tributaries which were disconnected from

the main river channel after the completion of river training in 1997. Some modified watercourses (W07, W14 and W15) are the road drainage channels, and others (W06, W13, W16 to W18) are the local storm drains. These watercourses converge with Kam Tin River before discharge to inner Deep Bay.

5.4.1.9 W24 is a semi-natural watercourse near Wah On Villa. It flows from east towards northwest via W23 and joins the road drains along Sam Tin Road (W19). It continuous its flows downstream through the modified watercourse (W21 and W22) and NTMDC to the west and eventually converges with Kam Tin River to the south.

5.4.1.10 There are natural (W36), semi-natural (W26, W28, W30 and W31) and modified (W25, W27, W29, W32 to W35) watercourses identified within the Ngau Tam Mei area. These watercourses generally converge with the NTMDC and eventually enter the Kam Tin River in southwest.

5.4.1.11 W38 and W39 are the semi-natural watercourses identified at the northwest of the Ngau Tam Shan. These watercourses flow towards north and converge with Shenzhen River in the north.

5.4.1.12 There are natural (W42), semi-natural (W40 and W41) and modified (W43) watercourses identified along the River Beas. W42 is the meander of old River Beas which is fragmented after the completion of river training in 2001. W40 and W41 run through Ki Lun Tsuen, flowing towards east, and converge with River Beas. W43 is the local drains which conveyed flow from Ki Lun Tsuen and Hang Tau Village. W45 is the downstream channel of Kwu Tung Reservoir which eventually joins River Beas near Kwu Tung.

5.4.1.13 W46 to W52 and W55 are modified watercourses identified at the northern boundary of the assessment area. W46 to W49 flow from east to west, converge with San Tin Eastern Main Drainage Channel and eventually enter the Shenzhen River to the north. W50 to W52 flow from west to east and joins River Beas near Kwu Tung. W55 flows from west to east, passes through Ma Tso Lung and converge with Ma Tso Lung Nullah near Ma Tso Lung Shun Yee Shan Tsuen.

5.4.1.14 Two major continuous pond area are identified within the assessment area, one is located adjacent to the Kam Tin River (P01 to P08), and the other one is located within Ngau Tam Mei (NTM) area (P17 to P20 and P20a). P01 and P02 are the continuous ponds at Nam Sang Wai and Tai Sang Wai which are also located within the WCA. P03 and P04 are ponds along the Yuen Long Bypass Floodway (W02), and P05 to P08 are ponds adjacent to the old Kam Tin River. According to the EIA reports for "*Northern Link*" (register no.: AEIAR-259/2024) and "*Development at Ngau Tam Mei Area*", most of the existing ponds within or partially within the NTM development area will be removed (P17) prior to the commencement of construction works of this Project. P18 and P19 are ponds scattered in NTM area outside the Project Site boundary. P20 and P20a are a continuous pond area located near the Ngau Tam Mei Water Treatment Works which will be enhanced its ecological values after completion of the Project.

5.4.1.15 P27 is Kwu Tung Reservoir, which was constructed to serve as an irrigation reservoir for the local farming community in the past. The original irrigation function of Kwu Tung Reservoir has diminished, and the water is now preserved for the use by Fire Services Department.

5.4.1.16 Other isolated ponds are scattered throughout the assessment area. P09 to P11 are ponds located in Sha Po, and P12 to P16 are ponds located near the area in Pok Wai, Man Yuen Chuen and San Wai Tsuen. P21 to P23 are ponds located at the south of San Tin which will be removed according to the EIA report for "*San Tin - Lok Ma Chau Development Node*" (register no.: AEIAR-261/2024). P24 to P26 and P24a are ponds scattered in Ki Lun Tsuen adjacent to River Beas. P28 is a pond located at downstream of Kwu Tung Reservoir and P29 is a pond in Tit Hang. Most of these ponds are located near village areas which observed to be managed by the nearby villagers.

5.4.1.17 The Project Site is located within the Wetland Conservation Area (WCA) and Wetland Buffer Area (WBA).

5.4.1.18 There are eight patches of conservation areas located within the assessment area. They are designated as conservation area under the Approved Kam Tin South Outline Zoning Plan (OZP) No. S/YL-KTS/15, Approved Kam Tin North OZP No. S/YL-KTN/11, Approved San Tin Technopole OZP No. S/STT/2, Draft Nam Sang Wai OZP No. S/YL-NSW/11 and Draft Ngau Tam Mei OZP No. S/YL-NTM/15.

5.4.1.19 West Rail Compensatory Wetlands is a constructed and managed wetland habitat, which was created at west of Kam Hing Wai since 2022, covers an area of approximately 11.27 ha to mitigate for the loss of wetlands in the Kam Tin Valley due to the construction of West Rail in accordance with Environmental Permit No. FEP-24/004/1998/J. This wetland habitat is currently maintained and managed by MTR Corporation.

5.4.1.20 Yuen Long Bypass Floodway Engineered Wetland (YLBF EW) is located near the downstream of YLBF (W02) and managed by AFCD since December 2006. It is an ecological mitigation measures to compensate for the ecological loss of fishponds and agricultural land due to construction of YLBF. This 7-hectare engineered wetland mainly comprises three main ponds, one seasonal shallow pond and one permanent shallow pond. There are also crushed brick and oyster shell ponds, reed beds and a deep-water pond within the wetland. The dry weather flow of the YLBF, after purified by passing through the oyster shells, crushed bricks and reed bed, flows into the engineered wetland to offer an ecologically enriched habitat for the wild birds, amphibians and dragonflies. It should be noted that this Project would not affect the water flows feeding to YLBF EW. No adverse water quality impact is therefore expected.

5.4.1.21 Lam Tsuen Country Park (LTCP) is located partially within the southern end of the assessment area, north of Kai Kung Leng. The nearest point of the Project Site is located adjacent to the LTCP near Wah On Villa. Designated in 1979, LTCP commands a total area of 1,520 ha in the northern New Territories that spans over Tai Po, Fanling and Yuen Long. The park is divided into two parts by Fan Kam Road: Tai To Yan and Kai Kung Leng.

## 5.5 Assessment Methodology

5.5.1.1 The assessment approach employed to assess the potential water quality impacts associated with the construction and operation of the Project followed the detailed technical requirements given in Appendix D of the Study Brief and the Project information as presented in **Section 2**. Criteria and guidelines for evaluating and assessing water pollution followed Annex 6 and 14 of the EIAO-TM.

5.5.1.2 The Project falls within the Deep Bay WCZ as designated under the WPCO (Cap. 358). The assessment area should include areas within 500m from the boundary of the Project and its associated works as identified in the EIA Study. The WSRs in the vicinity of the Project have been identified. Potential sources of water quality impact that may arise during the construction and operation stages of the Project were described, including point discharges and non-point sources to surface water runoff, sewage from workforce and polluted discharge generated from the Project. All the identified sources of potential water quality impact at the identified WSRs were evaluated and their impact significance determined. Practical water pollution control measures were recommended to mitigate identified water quality impacts.

## 5.6 Identification, Prediction and Evaluation of Environmental Impacts

### 5.6.1 Construction Phase

5.6.1.1 General construction works for the Project would be land-based only and within the Project Site. The potential sources of water quality impact associated with the land-based works would include:

- Construction works in Kam Tin River;
- Wastewater from general construction activities;
- Construction site run-off;
- Accidental spillage of chemicals;
- Sewage from construction workforce;
- Construction works near watercourses;
- Removal / diversion of watercourses;
- Removal / filling of ponds;
- Groundwater infiltration arising from tunnel construction; and
- Groundwater from contaminated areas, contaminated site run-off and wastewater from land decontamination.

*Construction works in Kam Tin River*

5.6.1.2 The construction of pier and viaduct structure across Kam Tin River would require the installation of new piers in Kam Tin River using non-dredged method to minimise the disturbance to the riverbed. Cofferdams will be erected at Kam Tin River and thereafter dewatering will be conducted inside the cofferdam to create a dry working zone for the construction. Piles will then be constructed within the watertight casing with piling rigs on barge or temporary steel platforms. The excavation and removal of materials from pile shaft will be carried out within pile casing to minimise water quality impact. Upon the completion of pile cap construction, the cofferdams will be removed to maintain the water flow within Kam Tin River. Adoption of the proposed non-dredged construction method not only enables safe and secured working area on water, but also provides cleaner, dry and tidier working environment.

5.6.1.3 The pier construction should be conducted by phases to avoid adverse hydrodynamic impact due to flow blockage increase during the interim construction stages. Since the cross-section area of the river channel to be blocked by the pier installation was anticipated localized and temporary, the change in flow regime

would be localized and any local water quality impact would expect to be transient. As a result, the potential impact on water quality and hydrology of Kam Tin River would be insignificant, no adverse water quality impact would be expected if the mitigation measures detailed in **Sections 5.7.1.1 to 5.7.1.4** are implemented properly.

*Construction Site Runoff and Wastewater from General Construction Activities*

5.6.1.4 Wastewater generated from construction activities, including general cleaning and polishing, wheel washing, dust suppression and utility installation may contain high SS concentrations and a certain amount of grease and oil. Wastewater may also be generated from the accumulation of solid waste such as debris, rubbish, plastic package and construction materials. Sediment laden run-off during site formation works, if uncontrolled, may carry pollutants (absorbed onto the particle surfaces) into the nearby water bodies. Potential pollution sources of site run-off may include:

- Run-off and erosion of exposed bare soil and earth, drainage channel, earth working area and stockpiles;
- Release of any bentonite slurries, concrete washings and other grouting materials with construction run-off or storm water;
- Wash water from dust suppression sprays and wheel washing facilities; and
- Fuel, oil and lubricants from maintenance of construction vehicles and equipment.

5.6.1.5 During rainstorms, site run-off would wash away the soil particles on unpaved lands and areas with the topsoil exposed. The run-off is generally characterized by high concentration of SS. Release of uncontrolled site run-off would increase the SS levels and turbidity in the nearby water environment. Site run-off may also wash away soil particles that were contaminated by the construction activities and therefore cause water pollution.

5.6.1.6 Wind-blown dust would be generated from exposed soil surfaces in works areas. It is possible that wind-blown dust would fall directly onto the nearby water bodies when a strong wind occurs. Dispersion of dust within the works areas may increase the SS levels in surface run-off causing a potential impact to the nearby sensitive receivers.

5.6.1.7 It is important that proper site practice and good site management be followed to prevent run-off with high level of SS from entering the surrounding waters. Best Management Practices (BMPs) in controlling construction site discharges are recommended for this Project as described in **Sections 5.7.1.5 to 5.7.1.20**. In view of the sensitive nature of the Deep Bay WCZ, an Emergency Response Plan (ERP) should also be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. An outline of the ERP is provided in **Sections 5.7.1.21 to 5.7.1.22**. With the implementation of BMPs to control construction site effluents and ERP to minimise water quality impact under emergency situations, disturbance of water bodies would be avoided and deterioration in water quality would be minimal.

*Accidental Spillage of Chemicals*

5.6.1.8 The use of engine oil and lubricants and their storage as waste materials have the potential to create impacts on the water quality if spillage occurs. Waste oil may infiltrate into the surface soil layer, or run-off into adjacent waterbodies, increasing hydrocarbon levels. Groundwater pollution may also be arisen from the improper use and storage of chemical and petroleum products within the site areas where groundwater infiltrates into. Infiltration of groundwater may occur at areas where there are faults and/or fissures in the rock mass. Nevertheless, the potential impacts could be mitigated by practical mitigation measures and good site practices as described in **Sections 5.7.1.23 to 5.7.1.25**.

*Sewage from Construction Workforce*

5.6.1.9 During the construction of the Project, the workforce on site will generate sewage effluents, which are characterized by high level of BOD, ammonia and *E. coli*. According to Table T-2 of "*Guidelines for Estimating Sewage Flows for Sewage Infrastructure Planning*" (GESF), the unit flow factor (UFF) is 0.23 $m^3$/day/employee (0.08 $m^3$/day/employee for commercial employee and 0.15 $m^3$/day/employee for employees of construction activities). For every 100 construction workers working simultaneously at the construction site, about 23 $m^3$ of sewage would be generated per day. Potential water quality impacts upon the local drainage and freshwater system may arise from this sewage effluent, if uncontrolled.

5.6.1.10 However, this temporary sewage can be adequately handled by interim sanitary facilities, such as portable chemical toilets. According to Section 2.2 of "*Reference Materials on Construction Site Facilities for Workers*" that issued by the Construction Industry Council (CIC), the number of toilet facilities provided on site shall be at a ratio of not less than 1 for every 30 workers. The number of the chemical toilets required for the construction site should be subject to later detailed design, the capacity of the chemical toilets, and contractor's site practices. A licensed contractor should be employed to provide appropriate and adequate portable toilets and be responsible for appropriate disposal and maintenance.

5.6.1.11 Provided that sewage is not discharged directly into stormwater drains or inland waters adjacent to the construction site, and temporary sanitary facilities are used and properly maintained, no adverse water quality impact would be anticipated. Mitigation measures and good site practices as given in **Sections 5.7.1.26 to 5.7.1.27** should be implemented.

*Construction Works near Watercourses*

5.6.1.12 Watercourses were identified within the Site as identified in **Section 5.4**. Construction works in close proximity to the inland watercourses may pollute the inland waters due to potential release of construction wastes and wastewater. Construction wastes and wastewater are generally characterised by high concentration of SS and elevated pH. Adoption of good housekeeping and mitigation measures would reduce the generation of construction wastes and potential water pollution. The implementation of measures to control run-off and drainage water will be important for the construction works adjacent to the inland water in order to prevent run-off and drainage water with high levels of SS from entering the water environment. With the implementation of BMPs as described in ProPECC PN 2/24 "*Construction Site Drainage*" and provision of mitigation measures as specified in ETWB TC (Works) No. 5/2005 "*Protection of natural*

*streams / rivers from adverse impacts arising from construction works*" (also see **Section 5.7.1.28**), it is anticipated that water quality impacts would be minimised.

*Removal / Diversion of Watercourses*

5.6.1.13 Construction of portals at Ngau Tam Shan (near the proposed Ngau Tam Mei Interchange) and Hadden Hill (near the Kwu Tung Reservoir), T-junction in Ki Lun Tsuen (east of Saddle Pass), as well as site formation works of Ngau Tam Mei Interchange will be required for construction of the Project. These construction works will remove/divert the existing watercourses (i.e. partial sections of W27, W31, W37 and W44 within the Project Site) from the existing route to the new drainage system. Removal of these existing watercourses would involve diversion of water flow from their existing routes to the new routes through the proposed covered drainage system and demolition of the existing watercourses. Construction of the new channel and new drainage network may involve excavation and use of concrete.

5.6.1.14 Potential impacts may be generated by discharge of concrete slurry and other grouting materials generated by concreting works as well as the release of construction materials, wastewater, excavated sediment, spillage and contaminants to the receiving waters in the downstream (due to soil excavation for construction of new drainage and demolition of the existing watercourses). All these construction works should be undertaken in dry conditions to avoid potential water quality impacts upon the downstream water quality.

5.6.1.15 The tentative works sequence for diversion and removal of watercourses is described below. Permanent or temporary drainage at the most downstream should first be established to intercept and divert the upstream flow. Construction of permanent or temporary drainage should be undertaken in a dry condition prior to receiving any water flow. Where necessary, precautionary measures in Appendix D of ETWB TC No. 5/2005, such as temporary isolation to other connected watercourse using sandbags and silt curtains, should be used to isolate the works areas from neighbouring waters. The permanent or temporary drainage should be constructed and completed before dewatering of the existing watercourse. Dewatering of watercourse should be performed by diverting the water flow to new or temporary drainage. Silt removal facilities should be adopted to treat the wastewater from dewatering operations prior to discharge. Construction works at watercourse should be undertaken only after flow diversion or dewatering operation is fully completed to avoid water flow in the works area. After completion of the construction works, the works area shall be cleaned up before receiving any water flow or connecting to any existing watercourse.

5.6.1.16 Excavated materials (including sediment) may be generated from the removal and diversion of watercourses. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.7.1.34** should be followed to minimise the potential environmental impacts arising from the excavated materials.

5.6.1.17 Mitigation measures for protection of downstream water quality from diversion and/or removal of watercourses are described in **Sections 5.7.1.29 to 5.7.1.32**. With adoption of the recommended mitigation measures, there should not be significant sediment and contaminant release to the downstream water.

*Removal / Filling of Ponds*

5.6.1.18 Due to construction of the proposed tunnel portal at Ngau Tam Shan and T-junction in Ki Lun Tsuen, the existing ponds near Ngau Tam Mei Water Treatment Works (P20a) will be reconstructed to a new wetland enhancement area (after complete construction of the Project as recommended in **Section 8**) and at south of Ki Lun Tsuen Playground (P24a) will be removed under this Project, while other ponds (including P06, P17, and P21 to P23) will be removed by other projects prior to commencement of construction of this Project.

5.6.1.19 All the affected ponds should be isolated and not connected to any existing watercourse. The associated construction works would include draining the water in ponds before filling up these areas or before commencement of any excavation and construction works. The water of these ponds to be drained would probably be sediment-laden and would carry a certain level of pollutants. Direct dumping of these drained waters to the nearby watercourse will not be allowed.

5.6.1.20 The drained water generated from dewatering of the ponds to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as practicable and any surplus water should be tankered away for disposal. Any surplus water should be treated as necessary before disposal in compliance with the WPCO and the TM-DSS. In order to minimise the potential impact, dewatering works at ponds and wet areas should be conducted within dry season to minimise the quantity of drained water. No direct discharge of drained water from these construction works will be allowed.

5.6.1.21 Excavated materials (including sediment) may be generated from the construction works in ponds. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.7.1.34** should be followed to minimise the potential water quality impacts arising from the excavated material. Together with the adoption of the mitigation measures for removal of ponds as recommended in **Section 5.7.1.33**, no adverse water quality impact would be expected.

*Groundwater Infiltration arising from Tunnel Construction*

5.6.1.22 Tunnels will be constructed at the sections passing through Ngau Tam Shan and Hadden Hill using drill and blast method which may result in infiltration of groundwater. The major concern from tunnel construction activities would be the increase in site runoff and the associated potential drawdown of groundwater in any soil and aquifer layers. Groundwater infiltration would affect the construction works and infiltrated water would carry away silt and other contaminants from site into the site drainage. Consideration should be taken at the early design stage to minimize the infiltration of groundwater. Underground development may also drawdown groundwater in any soil and aquifer layers if uncontrolled. Practical groundwater control measures (as given in **Sections 5.7.1.35 to 5.7.1.38**) should be followed to minimize the potential groundwater infiltration.

*Groundwater from Contaminated Areas, Contaminated Site Runoff and Wastewater from Land Contamination*

5.6.1.23 Potential land contamination issues have been identified within the Project Site. Proper land contamination remediation and mitigation measures are proposed in **Section 7**. Any contaminated material disturbed, or material which comes into

contact with the contaminated material, has the potential to be washed with site runoff into drainage system. Excavated contaminated materials would be properly stored, housed and covered to avoid generation of contaminated runoff. Open stockpiling of contaminated materials will not be allowed. Any contaminated site runoff will be properly treated and disposed in compliance with the requirements of the TM-DSS. Mitigation measures for contaminated site runoff are recommended under **Section 5.7.1.39**.

5.6.1.24 Groundwater pumped out or from dewatering process during excavation works in the contaminated areas might be potentially contaminated. Any contaminated groundwater will be either properly treated or properly recharged into the ground in compliance with the requirements of the TM-DSS. No direct discharge of contaminated groundwater will be adopted. Mitigation measures and monitoring requirements for contaminated groundwater discharge/recharge are recommended under **Sections 5.7.1.40 to 5.7.1.41**. With proper implementation of the recommended mitigation measures, no unacceptable water quality would be expected from the groundwater generated from contaminated areas and contaminated site runoff.

5.6.2 Operation Phase

5.6.2.1 Potential water quality impacts associated with the operation phase would include:

- Hydrodynamic and water quality impacts on Kam Tin River;
- Non-point source surface runoff from new impervious areas; and
- Sewage effluent generated from the new administration building.

*Hydrodynamic and Water Quality Impacts on Kam Tin River*

5.6.2.2 The permanent bridge pier structure in Kam Tin River has potential to obstruct the river flow. However, considering that the proposed bridge piers will be oriented with the existing bridge piers of Tsing Long Highway / Yuen Long Highway, reduction in overall cross-section area of Kam Tin River due to the presence of the existing and proposed piers is expected to be minimal. A separate Drainage Impact Assessment (DIA) under this Project has been conducted and concluded that the potential hydraulic impact to the drainage system would be minimal. No adverse hydrodynamic impact is therefore anticipated. No significant change in water quality regime, which associated with the hydrodynamic impact, is anticipated. No adverse hydrodynamic and water quality impacts would therefore be expected.

*Surface Runoff*

5.6.2.3 Surface runoff to be generated from the Project site is known as non-point source pollution. Any increase in paved area within the Project site will increase the chance of surface runoff. The presence of oil, grease and grit on surfaces of road within the Project site could be washed into the nearby drainage system or even into the watercourses during rainfall event. Surface runoff generated from the paved or developed areas may contain debris, refuse and dust. Practices that cleaning agents used for road washing may also affect the water quality of the nearby receiving water.

5.6.2.4 To minimise the impact from road runoff, all the road works planned under the Project should be designed with adequate drainage system and appropriate oil

interceptors, as required. The design of stormwater drains shall follow the relevant guidelines and practices as given in the ProPECC PN 1/23 "*Drainage Plans subject to Comment by the EPD*". With implementation of recommended measures as described in **Sections 5.7.2.2 to 5.7.2.10**, the potential water quality impact due to surface runoff is expected to be minor and no adverse water quality impact from non-point source surface run-off is expected.

*Sewage Effluent from New Administration Building*

5.6.2.5 Sewage generated from the operation staff of the proposed administration building (and portal plant building / plant room) located to the South of Ngau Tam Mei under the operation phase is characterized by high level of BOD, ammonia and *E. coli*. It would not cause adverse water quality impacts if the sewage would be properly treated and disposed.

5.6.2.6 All sewage generated from the operation staff would be collected and discharged to the public sewerage system and diverted to sewage treatment works for further treatments. No adverse water quality impact is therefore expected.

## 5.7 Mitigation of Adverse Environmental Impacts

5.7.1 Construction Phase

*Construction works in Kam Tin River*

5.7.1.1 The pilling works should be conducted by phases to avoid adverse hydrodynamic impact due to flow blockage increase during the interim construction stages. The method and sequence of the proposed pier works in Kam Tin River should be carefully designed so that wastewater and sediment laden water generated from the pilling works would be confined and physically separated from the watercourse.

5.7.1.2 All pilling, the associated excavation works and construction of pile caps in river should be fully enclosed by the cofferdam / watertight pile casing. Cofferdam and watertight pile casing should be constructed to isolate the construction activities from the river water. The detail design of cofferdams and watertight pile casing will be conducted by the Contractor during the construction phase to fulfil the requirements in DSD Technical Circular No. 1/2017 "*Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System*" for DSD approval in order to formulate feasible options of these temporary structures.

5.7.1.3 Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdam and watertight pile casing. The collected construction site surface runoff and ingress / seepage water should be diverted to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharged. Discharge licence issued by the EPD for discharging effluent from the construction site under the WPCO is needed. The discharge quality and quantity must meet the requirements specified in the discharge licence and follow the TM-DSS.

5.7.1.4 To further minimize any adverse water quality impact during the pilling and excavation works, silt curtains should be deployed to completely enclose the cofferdam / watertight pile casing prior to setting up piling works and installation of cofferdam / watertight pile casing. Silt curtains should only be removed after completion of pilling works and removal of cofferdam / watertight pile casing. The

Contractor should be responsible for the design, installation and maintenance of the silt curtain to minimize the impacts on water quality. The design and specification of the silt curtains should be submitted by the Contractor to the Project Engineer for approval.

*Construction Site Runoff and Wastewater from General Construction Activities*

5.7.1.5 Control of potential pollution of nearby water bodies during the construction phase of the Project should be achieved by measures to:

- prevent or minimise the likelihood of pollutants (generated from construction activities) being in contact with rainfall or run-off; and
- abate pollutants in the stormwater surface run-off prior to the discharge of surface run-off to the nearby water bodies.

5.7.1.6 These principle objectives should be achieved by implementation of the Best Management Practices (BMPs) of mitigation measures in controlling water pollution. The guidelines for handling and disposal of construction site discharges as detailed in the ProPECC PN 2/24 "*Construction Site Drainage*" should be followed, where applicable. All effluent discharged from the construction site should comply with the standards stipulated in the discharge licence under the WPCO. The following measures are recommended to protect water quality of the inland and coastal waters, and when properly implemented should be sufficient to adequately control site discharges so as to avoid water quality impacts.

*Construction Site Runoff*

5.7.1.7 Surface run-off from construction site should be discharged into storm water drains via adequately designed on-site wastewater treatment facility such as sand traps, silt traps, sedimentation tanks and sediment basins. Channels, earth bunds or sand bag barriers should be provided on site to properly direct surface run-off to such silt removal facilities. Perimeter channels at site boundaries should be provided where necessary to intercept surface run-off from outside the site so that it will not wash across the site. Catchpits and perimeter channels should be constructed in advance of site formation works and earthworks.

5.7.1.8 Silt removal facilities, channels and manholes should be maintained and the deposited silt and grit should be removed regularly, at the onset of and after rainstorm to ensure that these facilities are functioning properly at all times. Before disposal at the public fill reception facilities, the deposited silt and grit should be solicited in such a way that it can be contained and delivered by dump truck instead of tanker truck. Any practical options for the diversion and realignment of drainage should comply with both engineering and environmental requirements in order to provide adequate hydraulic capacity of all drains.

5.7.1.9 Construction works should be programmed to minimise soil excavation in the rainy seasons (generally from April to September). If soil excavation cannot be avoided in these months or at any time of year when rainstorms are likely, for the purpose of preventing soil erosion, temporarily exposed slope surfaces should be covered (e.g. by tarpaulin), and temporary access roads should be protected by crushed stone or gravel, as excavation proceeds. Intercepting channels should be provided (e.g. along the crest / edge of excavation) to prevent surface run-off from washing across exposed soil surfaces. Arrangements should always be in place to ensure

that adequate surface protection measures can be safely carried out well before the arrival of rainstorm.

5.7.1.10 Earthworks final surfaces should be well compacted and the subsequent permanent work or surface protection works should be carried out immediately after the final surfaces are formed to prevent erosion caused by rainstorms. Appropriate drainage like intercepting channels should be provided where necessary.

5.7.1.11 Measures should be taken to minimise the ingress of rainwater into trenches. If excavation of trenches in the rainy season is necessary, they should be dug and backfilled in short sections of length. Rainwater pumped out from trenches or foundation excavations should be discharged into storm drains via silt removal facilities.

5.7.1.12 Open stockpiles of construction materials (e.g. aggregates, sand and fill material) on sites should be covered with tarpaulin or similar fabric during rainstorms. Measures should be taken to prevent the washing away construction materials, soil, silt or debris into any drainage system.

5.7.1.13 Manholes (including newly constructed ones) should always be adequately covered and temporarily sealed so as to prevent silt, construction materials or debris from getting into the drainage system, and to prevent storm run-off from getting into foul sewers. Discharge of surface run-off into foul sewers must always be prevented in order not to unduly overload the foul sewerage system.

*Boring and Drilling Water*

5.7.1.14 Water used in ground boring and drilling for site investigation or rock/soil anchoring should as far as practicable be re-circulated after sedimentation. When there is a need for final disposal, the wastewater should be discharged into storm drains via silt removal facilities.

*Bentonite Slurries*

5.7.1.15 Bentonite slurries used in bored-pile construction should be reconditioned and reused wherever practicable. If the disposal of a certain residual quantity cannot be avoided, used bentonite slurry (if mixed with only inert fill materials) should be dewatered, and disposed of at a (i) public fill reception facility / area; or (ii) marine dumping ground (as the last resort) subject to obtaining a marine dumping licence from the Environmental Protection Department (EPD) on a case-by-case basis.

5.7.1.16 The water generated from the dewatering process should be treated to the respective effluent standards applicable to foul sewers, storm water drains or the receiving waters as set out in the TM-DSS issued under the WPCO.

*Wheel Washing Water*

5.7.1.17 All vehicles and plants should be cleaned before they leave a construction site to ensure no earth, mud, debris and the like is deposited by them on roads. A wheel washing bay should be provided at every site exit if practicable and wash-water should have sand and silt settled out or removed before discharging into storm drains. The section of construction road between the wheel washing bay and the public road should be paved to reduce vehicle tracking of soil and to prevent site run-off from entering public road drains.

*Acid Cleaning, Etching and Pickling Wastewater*

5.7.1.18 Acidic wastewater generated from acid cleaning, etching, pickling and similar activities should be neutralised to within the pH range of 6 to 10 before discharging into foul sewers. If there is no public foul sewer in the vicinity, the neutralized wastewater should be tankered off site for disposal into foul sewers or treated to a standard acceptable for discharge into storm water drains and the receiving waters.

*Rubbish and Litter*

5.7.1.19 Good site practices should be adopted to remove rubbish and litter from construction site so as to prevent the rubbish and litter from spreading from the site area. The construction site should be cleaned on a regular basis.

*Effluent Discharge*

5.7.1.20 There is a need to apply to EPD for a discharge licence for discharge of effluent from the construction site under the WPCO. The discharge quality must meet the requirements specified in the discharge licence. All the runoff and wastewater generated from the works areas should be treated so that it satisfies all the standards listed in the discharge licence under the WPCO. The beneficial uses of the treated effluent for other on-site activities such as dust suppression, wheel washing and general cleaning etc., can minimise water consumption and reduce the effluent discharge volume. If monitoring of the treated effluent quality from the works areas is required during the construction phase of the Project, the monitoring should be carried out in accordance with the relevant WPCO licence.

*Emergency Response Plan for Construction Site Discharges*

5.7.1.21 The following measures should be implemented by the Contractors to minimise the chance of emergency construction site discharge (due to failure of treatment facilities such as sand traps, silt traps, sedimentation basins, oil interceptors etc.):

- Provide spare or standby treatment facilities of suitable capacities for emergency replacement in case damage or defect or malfunctioning of the duty treatment facilities is observed.
- Conduct daily integrity checking of the construction site drainage and treatment facilities to inspect malfunctions, in particular before, during and after a storm event.
- Carry out regular maintenance or desilting works to maintain effectiveness of the construction site drainage and treatment facilities in particular before, during and after a storm event.

5.7.1.22 An ERP should be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. The ERP should give the emergency contacts to mobilise flood retention facilities and stakeholders to be notified as well as the details of the proposed construction site drainage system and the design and operation of duty and standby treatment facilities. The ERP should also provide the procedures and guidelines for routine integrity checking and maintenance of the drainage system and treatment facilities as well as the emergency response and rectification procedures to restore normal operation of the treatment facilities in case of treatment failure during emergency situation or inclement weather. The BMPs in

controlling water pollution arising from the construction activities and an event and action plan with action and limit levels for water quality monitoring should be included in the ERP. The ERP should be submitted to the Engineer for approval before commencement of the construction works.

*Accidental Spillage of Chemicals*

5.7.1.23 Contractor must register as a chemical waste producer if chemical wastes would be produced from the construction activities. The Waste Disposal Ordinance (Cap 354) and its subsidiary regulations in particular the Waste Disposal (Chemical Waste) (General) Regulation, should be observed and complied with for control of chemical wastes. The Contractor is also recommended to develop management procedures for chemicals used and prepare an emergency spillage handling procedure to deal with chemical spillage in case of accident occurs.

5.7.1.24 Any service shop and maintenance facilities should be located on hard standings within a bunded area, and sumps and oil interceptors should be provided. Maintenance of vehicles and equipment involving activities with potential for leakage and spillage should only be undertaken within the areas appropriately equipped to control these discharges.

5.7.1.25 Disposal of chemical wastes should be carried out in compliance with the Waste Disposal Ordinance. The Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes published under the Waste Disposal Ordinance details the requirements to deal with chemical wastes. General requirements are given as follows:

- Suitable containers should be used to hold the chemical wastes to avoid leakage or spillage during storage, handling and transport.
- Chemical waste containers should be suitably labelled, to notify and warn the personnel who are handling the wastes, to avoid accidents.
- Storage area should be selected at a safe location on site and adequate space should be allocated to the storage area.

*Sewage from Construction Workforce*

5.7.1.26 No discharge of sewage to the stormwater drains or inland water will be allowed. Adequate and sufficient portable chemical toilets should be provided in the works areas to handle sewage from construction workforce. A licensed collector should be employed to clean and maintain the chemical toilets on a regular basis.

5.7.1.27 Notices should be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the surrounding environment. Regular environmental audit of the construction site should be conducted to provide an effective control of any malpractices and achieve continual improvement of environmental performance on site.

*Construction Works near Watercourses*

5.7.1.28 To minimise the potential water quality impacts from the construction works located near any inland watercourses, the practices outlined in ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from*

*construction works*" should be adopted where applicable. Relevant mitigation measures are listed below:

- Impermeable sheet piles and / or cofferdams should be used as required to divert water flow from the construction works area so that all the construction works would be undertaken within a dry zone and physically separated from the watercourses.
- Construction works close to the inland watercourses should be carried out in dry season as far as practicable where the flow in the surface channel/stream is low.
- The use of less or smaller construction plants may be specified in works areas close to the inland watercourses.
- Temporary storage of materials (e.g. equipment, filling materials, chemicals and fuel) and temporary stockpile of construction materials should be located well away from any watercourses during carrying out of the construction works.
- Stockpiling of construction materials and dusty materials should be covered and located away from the any watercourses.
- Construction debris and spoil should be covered up and/or disposed of as soon as possible to avoid being washed into the nearby water receivers.
- Construction activities, which generate large amount of wastewater, should be carried out at a distance away from the watercourses, where practicable.
- Mitigation measures to control site run-off from entering the nearby water environment should be implemented to minimise water quality impacts. Surface channels should be provided along the edge of the waterfront within the work sites to intercept the run-off.
- Construction effluent, site run-off and sewage should be properly collected and/or treated.
- Any temporary works site inside the stormwater watercourses should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the stormwater quality.
- Proper shoring may need to be erected in order to prevent soil/mud from slipping into the inland watercourses.

*Removal / Diversion of Watercourses*

5.7.1.29 The construction method and sequence of diversion and removal of watercourse should be carefully designed so that all the construction works including any excavation would be undertaken within a dry zone and physically separated from the watercourses downstream.

5.7.1.30 The tentative works sequence for diversion and removal of watercourse should be duly planned to minimize potential water quality impacts as described below. Permanent or temporary drainage at most downstream should first be established to intercept and divert the upstream flow. Construction of permanent or temporary drainage should be undertaken in a dry condition prior to receiving any water flow. Where necessary, precautionary measures in Appendix D of ETWB TC No. 5/2005, such as temporary isolation to other connected watercourse using sandbags and

silt curtains, should be used to isolate the works areas from neighbouring waters. The permanent or temporary drainage should be constructed and completed before dewatering of the existing watercourse. Dewatering of watercourse should be performed by diverting the water flow to new or temporary drainage. Silt removal facilities should be adopted to treat the wastewater from dewatering operations prior to discharge. Construction works at watercourse should be undertaken only after flow diversion or dewatering operation is fully completed to avoid water flow in the works area. After completion of the construction works, the works area shall be cleaned up before receiving any water flow or connecting to any existing watercourse.

5.7.1.31 The Contractor should provide a dry zone for all the construction works to be undertaken in watercourses and stormwater drainage following the tentative works sequence as described above or using other approved methods as appropriate to suit the works condition. The flow diversion works should be conducted in dry season, where possible, when the flow in the watercourse is low. The wastewater and ingress water from the site should be properly treated to comply with the requirements of the WPCO licence and the TM-DSS before discharge.

5.7.1.32 The site practices outlined in the ProPECC PN 2/24 "*Construction Site Drainage*" and ETWB TC (Works) No. 5/2005 "*Protection of natural streams/rivers from adverse impacts arising from construction works*" as recommended in **Sections 5.7.1.5** to **5.7.1.20** and **5.7.1.28** should also be followed to minimise the potential water quality impacts from construction in watercourses.

*Removal / Filling of Ponds*

5.7.1.33 Construction works at the existing affected ponds/wet areas should be conducted only after dewatering of these ponds is fully completed. The drained water generated from the dewatering of these ponds to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as possible. Any surplus drained water should be tankered away for disposal in a controlled manner. No direct discharge of drained water to the stormwater drainage system or marine water should be allowed.

*Disposal of Excavated Materials and Sediment*

5.7.1.34 All excavated materials generated from construction of viaduct across Kam Tin River, removal and diversion of watercourses, removal and construction works in ponds should be collected and handled in compliance with the Waste Disposal Ordinance. Excavated sediment, if any, generated from the excavation activities in the watercourses should be tested and classified in accordance with the Section 4.2.1 of Project Administration Handbook (PAH) for Civil Engineering Works "*Marine Management of Dredged / Excavated Sediment*" for determining the disposal arrangement for the sediment. The disposal of excavated sediments should be minimised according to the relevant requirements in **Section 6**. No direct disposal of the construction wastes or excavated materials into the stormwater drainage system and marine water would be allowed.

*Groundwater Infiltration arising from Tunnel Construction*

5.7.1.35 Appropriate measures should be implemented to minimize the groundwater infiltration during the tunnel construction. The groundwater control strategies include:

- Probing Ahead: As normal practice, the Contractor will undertake rigorous probing of the ground ahead of excavation works to identify zones of significant water inflow. The probe drilling results will be evaluated to determine specific grouting requirements in line with the tunnel advance. In such zones of significant water inflow that could occur as a result of discrete, permeable features, the intent would be to reduce overall inflow by means of cut-off grouting executed ahead of the tunnel advance.
- Pre-grouting: Where water inflow quantities are excessive, pre-grouting will be required to reduce the water inflow into the tunnel. The pre-grouting will be achieved via a systematic and carefully specified protocol of grouting.
- In principle, the grout pre-treatment would be designed on the basis of probe hole drilling ahead of the tunnel face.
- The installation of waterproof lining would also be adopted after the formation of the tunnels.

5.7.1.36 In the event of excessive groundwater infiltration being observed as a result of the tunnelling or excavation works even after incorporation of the water control strategies, post-grouting should be applied as far as practicable as described below:

- Post-grouting: Groundwater drawdown will be most likely due to inflows of water into the tunnel that have not been sufficiently controlled by the pre-grouting measures. Where this occurs, post grouting will be undertaken before the lining is casted. Whilst unlikely to be required in significant measure, such a contingency should be allowed for reduction in permeability of the tunnel surround (by grouting) to limit inflow to acceptable levels.

5.7.1.37 The practical groundwater control measures stated above are proven technologies and have been extensively applied in other past projects. These measures or other similar methods, as approved by the Engineer to suit the works condition shall be applied to minimize the groundwater infiltration. With implementation of the above mitigation measures, no adverse water quality impact from groundwater drawdown is expected.

5.7.1.38 In case seepage of groundwater occurs, groundwater should be pumped out from works areas and discharged to the drainage system via silt trap. Groundwater from dewatering process should also be discharged to the storm system via silt removal facilities.

*Groundwater from Contaminated Areas, Contaminated Site Runoff and Wastewater from Land Contamination*

5.7.1.39 Remediation of contaminated land should be properly conducted following the recommendations of Land Contamination Assessment in **Section 7**. Any excavated contaminated material and exposed contaminated surface should be properly housed and covered to avoid generation of contaminated runoff. Open stockpiling of contaminated materials should not be allowed. Any contaminated runoff or wastewater generated from the land decontamination processes should be properly collected and diverted to wastewater treatment facilities (WTF) as necessary. The WTF shall deploy suitable treatment processes (e.g. oil interceptor / activated carbon) to reduce the pollution level to an acceptable standard and remove any prohibited substances (such as total petroleum hydrocarbon) to an undetectable range. All treated effluent from the wastewater treatment system shall

meet the requirements as stated in TM-DSS and should be either discharged into the foul sewers or tankered away for proper disposal.

5.7.1.40 No direct discharge of groundwater from contaminated areas should be adopted. Prior to any excavation works within the potentially contaminated areas, the baseline groundwater quality in these areas should be reviewed based on the past relevant site investigation data and any additional groundwater quality measurements to be performed with reference to "*Guidance Note for Contaminated Land Assessment and Remediation*" and the review results should be submitted to EPD for examination. If the review results indicated that the groundwater to be generated from the excavation works would be contaminated, this contaminated groundwater should be either properly treated or properly recharged into the ground in compliance with the requirements of the TM-DSS. If wastewater treatment is to be deployed for treating the contaminated groundwater, the wastewater treatment unit shall deploy suitable treatment processes (e.g. oil interceptor / activated carbon) to reduce the pollution level to an acceptable standard and remove any prohibited substances (such as total petroleum hydrocarbon) to an undetectable range. All treated effluent from the wastewater treatment plant shall meet the requirements as stated in the TM-DSS and should be either discharged into the foul sewers or tankered away for proper disposal.

5.7.1.41 If deployment of wastewater treatment is not feasible for handling the contaminated groundwater, groundwater recharging wells should be installed as appropriate for recharging the contaminated groundwater back into the ground. The recharging wells should be selected at places where the groundwater quality will not be affected by the recharge operation as indicated in Section 2.3 of TM-DSS. The baseline groundwater quality should be determined prior to the selection of the recharge wells, and submit a working plan to EPD for agreement. Pollution levels of groundwater to be recharged shall not be higher than pollutant levels of ambient groundwater at the recharge well. Groundwater monitoring wells should be installed near the recharge points to monitor the effectiveness of the recharge wells and to ensure that no likelihood of increase of groundwater level and transfer of pollutants beyond the site boundary. Prior to recharge, free products should be removed as necessary by installing the petrol interceptor. The Contractor should apply for a discharge licence under the WPCO through the Regional Office of EPD for groundwater recharge operation or discharge of treated groundwater.

5.7.2 Operation Phase

*Hydrodynamic and Water Quality Impacts on Kam Tin River*

5.7.2.1 As discussed in **Section 5.6.2.2**, no significant change in flow regime and water quality associated with the presence of the bridge piers in Kam Tin River is anticipated. No adverse hydrodynamic and water quality impacts would be expected during the operation phase, and no mitigation measures are therefore considered necessary.

*Stormwater Management Practices and Stormwater Pollution Control Plan*

5.7.2.2 The design of site drainage should follow the guidelines and practices as given in the ProPECC PN 1/23 "*Drainage Plan subject to Comments by the EPD*". With reference to Clause 2.3, Annex 6 of the EIAO-TM, mitigation measures including stormwater management practices to reduce stormwater pollution arising from the Project are as follows.

*Design Measures to Control Erosion and Run-off Quantity*

5.7.2.3 Exposed surface shall be avoided within the Project Site to minimise soil erosion. The Project Site shall be either hard paved or covered by landscaping area and plantation where appropriate.

5.7.2.4 The watercourses within the Project Site should be retained as far as practicable to maintain the original flow path. The drainage system should be designed to avoid flooding.

5.7.2.5 Green areas/tree/shrub planting etc. should be introduced within the Project Site as far as possible including open space and along roadside amenity strips and central dividers, which can help to reduce soil erosion.

*Devices / Facilities to Control Pollution*

5.7.2.6 Screening facilities such as standard gully grating and trash grille, with spacing which is capable of screening large substances such as fallen leaves and rubbish should be provided at the inlet of drainage system.

5.7.2.7 Road gullies with standard design and silt traps and oil interceptors should be incorporated during the detailed design to remove particles present in stormwater run-off, where appropriate.

5.7.2.8 Evergreen tree species, which in general generate relatively smaller amount of fallen leaves, should be selected where possible.

*Administrative Measures*

5.7.2.9 Good management measures such as regular cleaning and sweeping of road surface/open areas by the Food and Environmental Hygiene Department's contractor is suggested. The road surface/open area cleaning should also be carried out prior to occurrence of rainstorm.

5.7.2.10 Manholes, as well as storm water gullies, ditches provided among the Project Site should be regularly inspected and cleaned (e.g. monthly) by the Highways Department's contractor. Additional inspection and cleansing should be carried out before forecast heavy rainfall.

*Sewage Effluent from New Administration Building*

5.7.2.11 All sewage generated from the operation staff should be collected and discharged to the public sewerage system and diverted to sewage treatment works for further treatments. No adverse water quality impact is therefore expected.

## 5.8 Cumulative Impacts

### 5.8.1 Concurrent Projects

5.8.1.1 According to **Section 2**, construction of the Project commences tentatively in 2027 and completion in or before 2036. The Project will be constructed and operated with other concurrent projects, including "*San Tin Technopole*" (STT), "*Kwu Tung North New Development Area*" (KTN NDA), "*Ngau Tam Mei New Development Area*" (NTM NDA), "*Ngau Tam Mei Water Treatment Works Extension*" (NTM WTW

Extension), "*Northern Link*" (NOL), "*Northern Metropolis Highway - Kwu Tung Section*" (NMH-KT), "*Northern Metropolis Highway - Tin Shui Wai Section*" (NMH-TSW), planned residential development under Land Sharing Pilot Scheme Application no. LSPS/002, and planned developments under planning Application no. Y/YL-NSW/7, Y/YL-NSW/8, Y/YL-NSW/9, A/YL-NSW/274, A/YL-NSW/348, A/YL-NSW/349, and Y/NE-KTS/18. These projects are considered to have cumulative impact with the Project during operation phase. Some planned projects, including "*Proposed Nam Sang Wai Wetland Conservation Park*", "*Site Formation and infrastructure Works for Public Housing Development at Sha Po, Shap Pat Heung and Tai Kei Leng, Yuen Long*", and planned residential development under planning Application no. A/YL-NSW/274, A/YL-NSW/314, A/YL-NSW/293, A/YL-KTN/604, A/YL-KTN/663, and A/YL-KTN/964 are still under review by the relevant authorities when preparing this report. Cumulative operation impact with the Project has been considered for conservative approach.

### 5.8.2 Construction Phase

5.8.2.1 According to the EIA reports for "*San Tin / Lok Ma Chau Development Node*" (register no.: AEIAR-261/2024), "*North East New Territories New Development Areas*" (register no.: AEIAR-175/2013), "*Development at Ngau Tam Mei Area*" (register no.: EIA-316/2025), "*Ngau Tam Mei Area Water Treatment Works Extension*" (register no.: AEIAR-262/2024) and "*Northern Link*" (register no.: AEIAR-259/2024), potential sources of water quality impacts for construction of STT, KTN NDA, NTM NDA, NTM WTW Extension and NOL were identified to be land-based, including wastewater generated from general construction activities, construction site runoff, sewage from construction workforce, accidental spillage of chemicals, construction works near and/or in the watercourses, removal and/or diversion of watercourses and/or ponds, sterilisation of water mains prior to its commission, groundwater infiltration arising from tunnelling and underground works, groundwater from potential contaminated areas, contaminated site runoff and wastewater from land decontamination. Nevertheless, with implementation of the recommended mitigation measures as detailed in ProPECC PN 2/24 "*Construction Site Drainage*", ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*", BMPs and effective site management for individual projects, no adverse water quality impact is expected.

5.8.2.2 According to the project profiles for "*Northern Metropolis Highway - Tin Shui Wai Section*" (No. PP-691/2025) and "*Northern Metropolis Highway - Kwu Tung Section*" (No. PP-690/2025), potential sources of water quality impacts for construction of NMH-TSW and NMH-KT would be associated with wastewater generated from general construction activities, construction site runoff, sewage from construction workforce, accidental spillage of chemicals, construction works in Shan Pui River / Kam Tin River / Ng Tung River, construction works near the watercourses, diversion of watercourses and/or ponds (if necessary), groundwater infiltration arising from tunnelling works, groundwater from potential contaminated areas, contaminated site runoff and wastewater from land decontamination. With implementation of good site practices and mitigation measures including adoption of non-dredge method for construction of piles and piers in Shan Pui River / Kam Tin River / Ng Tung River, guidelines as given in ProPECC PN 2/24 "*Construction Site Drainage*" and ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*", potential water quality impact would be minimal. No adverse water quality impact is expected.

5.8.2.3 For planned residential developments (i.e. Application no.: LSPS/002, Y/YL-NSW/7, Y/YL-NSW/8 and Y/YL-NSW/9), potential sources of water quality construction impacts would be associated with wastewater generated from general construction activities and demolition works (for relocation of soy sauce factory), construction site runoff, sewage from construction workforce, accidental spillage of chemicals, and removal and/or modified of existing watercourses and ponds. With implementation of good site practices and mitigation measures as given in ProPECC PN 2/24 "*Construction Site Drainage*" and ETWB TC (Works) No. 5/2005 "*Protection of natural streams / rivers from adverse impacts arising from construction works*", potential water quality impact would be minimal. No adverse water quality impact is expected.

5.8.2.4 As no adverse water quality impacts are expected from the concurrent projects as discussed in **Sections 5.8.2.1 to 5.8.2.3** during construction phase, no adverse cumulative water quality impact during construction phase would therefore be expected.

5.8.3 Operation Phase

5.8.3.1 During operation phase, the development and infrastructural projects for STT, KTN NDA, NTM NDA, NTM WTW Extension, WCPS, NOL, NMH-TSW and NMH-KT, site formation and infrastructural works for public housing developments at Sha Po, Shap Pat Heung and Tai Kei Leng, and planned residential developments adjacent to the Project Site (Application no.: LSPS/002, Y/YL-NSW/7, Y/YL-NSW/8, Y/YL-NSW/9, A/YL-NSW/293, A/YL-NSW/314, A/YL-NSW/274, A/YL-NSW/293, A/YL-NSW/348, A/YL-NSW/349, A/YL-KTN/964, A/YL-KTN/604, A/YL-KTN/663, and Y/NE-KTS/18) are considered operating in conjunction with the Project. Key potential sources of water quality operation impacts would be related to discharge of sewage and wastewater from the developments and non-point source surface runoff from new road surfaces and development areas. Stormwater control measures including provision of adequate drainage system with suitable pollutant removal devices (i.e. silt traps and/or oil interceptors), blue-green infrastructure and stormwater management practices would be implemented to minimise the non-point source pollution. A new Effluent Polishing Plant (EPP) has been proposed under the EIA study "*San Tin / Lok Ma Chau Development Node*" (register no.: AEIAR-261/2024) to collect the sewage and wastewater generated from the new development area (i.e. STT, NTM NDA, WCPS, etc.). The EIA study "*North East New Territories New Development Areas*" (register no.: AEIAR-175/2013) proposed to upgrade the Shek Wu Hui Sewage Treatment Works to EPP (SWH EPP) with a design tertiary treatment capacity of 190,000 $m^3$/day to convey and treat the sewage generated from the KTN NDA and other adjacent areas. Since all sewage and wastewater generated from the existing and new development areas would be conveyed to the public sewerage system for treatment before discharge, and no direct discharge of sewage and wastewater to the watercourses would be allowed, no adverse water quality impact associated with sewage effluents from the developments is expected. No adverse cumulative water quality impact during operation phase would therefore be expected.

## 5.9 Evaluation of Residual Impacts

5.9.1.1 With proper implementation of the recommended mitigation measures, no residual water quality impact would be anticipated from the Project during construction and operation phases.

## 5.10 Environmental Monitoring and Audit

5.10.1.1 With full implementation of the mitigation measures such as good site practices, storm water pollution control plan, proper construction method, etc, no adverse impact is anticipated. No residual impact and cumulative impact are anticipated during both the construction and operation phases of the Project. In order to ensure effectiveness of the implemented mitigation measures, regular water quality monitoring in Kam Tin River is recommended during the construction phase. Details of the audit requirements are provided in the stand-alone EM&A Manual for the Project.

## 5.11 Conclusion

5.11.1.1 The potential water quality impacts associated with the land-based construction works include construction works in Kam Tin River, wastewater generated from general construction activities, construction site runoff, accidental spillage of chemicals, sewage from construction workforce, construction works near watercourses, removal and/or diversion of watercourses and/or ponds, groundwater infiltration arising from tunnel construction, groundwater from potential contaminated areas, contaminated site runoff and wastewater from land decontamination. Given these impacts could be mitigated and controlled by implementing the recommended mitigation measures, no adverse water quality impact during construction phase is expected. Regular site inspections and water quality monitoring should be undertaken routinely to inspect the construction activities and works area to ensure the recommended mitigation measures are proper implemented.

5.11.1.2 During operation phase, potential water quality impacts are related to surface runoff generated from the new paved areas as well as the change in flow regime and water quality impacts due to the presence of bridge piers in Kam Tin River. Provided that the recommended mitigation measures for the drainage system are properly implemented, the associated water quality impacts are expected to be minimal and acceptable. No significant change in flow regime and water quality in Kam Tin River is expected and hence, no adverse hydrodynamic and water quality impacts on Kam Tin River is expected. No adverse water quality impact during operation phase is therefore expected.